

SALTA Database

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1 Database

As noted towards the end of the chapter on SALTA (??), one of the main takeaways is how valuable user annotations, including perceptual cues, structural judgments, and interpretive decisions are, as proxies for the perceptual processes they encode. To make use of this resource, however, user-reviewed segmentation data must be stored in a format that is reproducible, long-lasting, and useful for future research. This led to the idea of developing a dedicated multimodal database which integrates the feature data together with the user annotations from a given performance. The segment annotations in that future database could then serve as ground truth data for the training of machine learning models.

1.1 Multimodal Corpora

Christodoulou et al. provide a framework that explains the challenges and best practices for the construction and evaluation of multimodal music datasets.¹ Their definition of multimodality as the intentional combination of different types of information, designed for specific analytical tasks, aligns closely with the design philosophy of SALTA, which integrates various sensory modalities including auditory, motion, and sensor data into one unified and synchronised temporal framework. The authors show that consistent normalisation of data, documentation, and handling missing data across modalities are important for data quality and interoperability.

Visi et al. highlight some important considerations for the modeling of multimodal corpora. They stress the use of various modalities (e.g., sound, motion, physiological data) in synchronised datasets, which allows a systematic exploration of cross-modal relations in expressive behaviour.² Visi et al. also found an important aspect called performer-informed or qualitative annotation, which contextualises computational features in the subjective and experiential domains of performance. While SALTA does not currently incorporate such qualitative annotation layers, its

¹ They emphasise methodological transparency, cross-modal alignment, and robust pre-processing procedures (Christodoulou et al., 2024).

² Visi et al. (2020).

modular and open architecture could accommodate them easily if a similar proposal came forward for future developments.

1.2 Format Choice

The choice of the format to store the segmentation and annotation data that are produced by the SALTA pipeline (described in ??) is not trivial. This storage needs to be done in adequate containers that ensure long-term accessibility and intelligibility, including 'enough descriptors and metadata to make the data sets clear and understandable for others'.³ In other words, structuring the data in ways that make it *queryable*, interoperable, and comprehensible to future researchers. The metadata should clarify the provenance, modality, and context of each dataset, without storing any sensitive information. We shall therefore favour data-exchange formats that balance openness, human readability, and archival durability. Table 1.1 summarises several formats currently under consideration, comparing them against criteria for suitability, interoperability, and long-term preservation.

XML vs JSON for Segment Annotation Formats

Feature	JSON	XML
Readability	More human-readable; concise and clean syntax	Verbose and harder to visually parse
Tool Support	Excellent; widely supported in modern software and web environments	Legacy-oriented; still supported in many tools
Schema Validation	JSON Schema is available but less strict	Robust; strong schema validation with XSD
Compatibility	Well-suited for modern APIs and languages	Strong with legacy systems and document-oriented formats
Extensibility	Flexible; easy to add new fields without breaking structure	Namespaces and schemas allow modular growth

³ Jensenius et al. (2007).

Best Fit for Segment Data	Well-aligned with structured time-aligned data like segments	Can be overly complex for simple time-label structures
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Table 1.2: Comparison of JSON and XML as file formats for segment-based annotations

Once Table 1.1 makes clear that the best candidate formats are XML and JSON, a closer look (Table 1.2) illustrates how JSON offers a more concise, readable, and structurally transparent way to represent segmented performance annotations. While XML allows for rigorous validation and integration with document-based workflows, it is more verbose and less convenient for handling lists of simple, time-aligned data objects like performance segments. JSON’s lightweight syntax and wide support across modern programming environments make it especially well-suited for projects that aim for flexibility, maintainability, and ease of use in tools for visualisation, analysis, or web integration. Since the use case involves evolving a format for segmentations of performance files—with likely goals such as extensibility, human-readability, and integration with modern pipelines—JSON provides the better balance of simplicity and scalability.

1.3 Taxonomy

An explicit requirement of our database is, of course, the ability to classify performance data from a wide range of artistic practices, be it dance, music, theatre, performance art, or any hybrid artistic forms. Here is where the need for a taxonomy of the performing arts is most evident. It should be coherent, interoperable and established in the field. Luckily, authoritative vocabularies already exist in cultural and academic circles.

The first major source is the Library of Congress Genre/Form Terms (LCGFT),⁴ a large, curated vocabulary that includes terminology for musical genres, theatrical forms, dance types, and interdisciplinary practices. Widely used in international cataloguing, LCGFT provides stable identifiers, a clear hierarchical structure, and high interpretability for both human and machine use. Complementing this, the Library of Congress Medium of Performance Thesaurus (LCMPT)⁵ offers an extremely detailed controlled vocabulary for musical performance modalities—including

⁴ LCGFT: <https://id.loc.gov/authorities/genreForms.html>, accessed 15 November 2025.
⁵ LCMPT: <https://id.loc.gov/authorities/performanceMediums.html>, accessed 15 November 2025.

instruments, vocal types, ensemble formations, and extended techniques. While LCMPT is music-specific, its precision is invaluable for annotating the many musical excerpts and performance experiments relevant to this research.

A second important pillar is the Getty Art & Architecture Thesaurus (AAT), an extensive ontology covering terms related to visual and performing arts, including performance genres, acting and staging conventions, dance forms, intermedia practices, and ritual performance.⁶ Getty AAT is particularly well suited for cross-disciplinary contexts because it provides broader conceptual categories and well-defined semantic relations—broader/narrower/related terms—that allow the system to capture both traditional and contemporary performance practices without implicitly prioritising any single discipline.

Given the relevance of dance and movement-based performance in many of the capture sessions carried out in the context of this dissertation, we also draw on the dance classification system used by the International Dance Council (CID), which includes academic, ethnic, social, contemporary, and improvisational dance forms. Although less formalised than LCGFT or AAT, CID's vocabulary remains one of the most internationally recognised frameworks specifically dedicated to dance and therefore enriches the system in areas where library-oriented metadata tends to be sparse.

Because this research is situated within a European institutional landscape, we also consider the Europeana Data Model (EDM) and its associated controlled vocabularies for cultural-heritage metadata.⁷ Europeana classifies performing-arts materials through a combination of Dublin Core terms, domain-specific refinements, and multilingual vocabularies used across European archives. Incorporating Europeana-aligned terminology strengthens the system's compatibility with European digital collections, which is relevant both to long-term sustainability and to future interoperability with cultural institutions inside the EU.

Taken together, these sources provide a taxonomy that is both broad (covering theatre, dance, music, performance art, intermedia, ritual performance, site-specific work, etc.) and precise (especially in musical and choreographic detail). The combined system benefits from the semantic clarity and stability of library science (LCGFT, LCMPT), the conceptual generality of cultural heritage ontologies (Getty AAT), and the culturally grounded specificity of European metadata frameworks (Europeana EDM).

⁶ AAT: <https://www.getty.edu/research/tools/vocabularies/aat/>, accessed 15 November 2025.

⁷ Europeana EDM: <https://pro.europeana.eu/page/edm-documentation>, accessed 15 November 2025.

Custom Format definition

The output format of the SALTA pipeline is documented and publicly accessible as a GitHub repository archived via Zenodo.⁸ The fields defined in that specification have been constructed according to the taxonomic sources mentioned earlier and include, besides the full formal definition,⁹ several example files across different capture modalities.

⁸ By making the format openly available and version-controlled, the repository ensures transparency, extensibility, and long-term sustainability, while also inviting further development and contribution from the wider research community. Repository DOI: <https://doi.org/10.5281/zenodo.17624041>.

⁹ Format definition: <https://github.com/AdrianArtacho/segmFORMAT>.

Format	Open / Standardised	Human- readable	Supports hierarchy	Long-term preservation	Compatibility / Ecosystem	Remarks and Typical Use
CSV (SALTA internal)	Yes (plain text)	Yes	No (flat structure)	Moderate (UTF-8 text; prone to schema drift)	Universal support across platforms; readable by SALTA, Python, Excel, R	Used internally for feature matrices and segmentation outputs; simple but lacks structure for multimodal metadata.
XML / segXML	Yes (XML-based, open schema)	Yes	Yes (nested elements)	High (validated XSD schema; archival stability)	Compatible with Sonic Visualiser, Praat, and other XML annotation tools	Extensible XML schema for multimodal segment annotations; hierarchical and self-descriptive.
JSON (CosmoNote style)	Yes (open)	Yes	Yes (nested key-value pairs)	Moderate (dependent on platform; good for exchange)	Ideal for web apps, REST APIs, and JavaScript environments; used by CosmoNote	Lightweight and human-readable; integrates easily with browser tools and Python; similar to CosmoNote.
MEI (Music Encoding Initiative)	Yes (community standard)	Partially (verbose XML)	Yes (deep hierarchical model)	High (archival-grade XML standard)	Supported by Verovio, MEI Viewer, and musicology software	Rich in musicological semantics; can embed structural segmentation events; heavy for multimodal use.
RDF / Linked Data	Yes (W3C standard)	No	Yes (semantic graph)	High (ontology-based, persistent URIs)	Interoperable with semantic-web frameworks; accessible via SPARQL, JSON-LD	Ideal for linking metadata and provenance; not suited for dense time-series data.
HDF5	Open binary format (HDF Group)	No	Yes (hierarchical groups)	High (scientific archival standard)	Used in scientific computing, TensorFlow, MATLAB, h5py	Efficient for large multimodal datasets and feature matrices; less transparent for manual editing.

Table 1.1: Comparison of candidate data-exchange formats for storing segmentation data in the SALTA framework.

2 Format Specification

Root-Level Keys

meta General metadata about the annotation and capture context.

salta Internal database information (SALTA system).

segmentations A dictionary of one or more segmentation entries.

meta

title (*string*) — Name of the segmentation dataset.

source_file (*string*) — Filename of the corresponding media.

duration (*float*) — Duration of the media file.

units (*string*) — Time unit (e.g., "seconds").

annotators (*array[string]*) — List of annotators.

version (*string*) — Format version.

date_capture (*string, optional*) — Date when media was recorded.

datum_capture (*string, optional*) — Duplicate or alternative capture date.

datum_entered (*string*) — Date the metadata was entered in DB.

comment (*string, optional*) — Free-text comment.

link (*string, optional*) — External project/media link.

modality (*object*) — Information about data modalities.

modality

VIDEO (*object, optional*)

one_take (*boolean, optional*)

single_subject (*boolean, optional*)

homogeneous_background (*boolean, optional*)

good_lighting (*boolean, optional*)

AUDIO (*object, optional*)

channels (*integer, optional*)

raw_preprocessed (*string, optional*) — e.g., "raw" or "preprocessed".

subjective_quality (*integer, 1–7, optional*)

IMU (*object, optional*)

type (*string, optional*) — e.g., "wearable".

device_model (*string, optional*)

num_sensors (*integer, optional*)

CUSTOM (*object, optional*)

description (*string, optional*) — Free description of this modality.

salta

project_id (*string*) — Unique project ID in SALTA.

primary_category (*string*) — Main category based on ontology.

secondary_category (*string, optional*) — Optional sub-category.

date_entered (*string*) — Date this data entered SALTA.

uploader_id (*string*) — User who submitted the data.

segmentations

segmentationX (*object*) — Segmentation entry (segmentation0, segmentation1, ...)

type (*string*) — One of: "AUTO", "weighted", "annotated".

date (*string*) — Date of segmentation.

user_feedback (*integer, optional*) — User satisfaction rating (e.g., 1–7).

segments (*array*) — List of segments (see below).

segments

Each segment is a dictionary with:

start (*float*) — Start time (in seconds).

end (*float*) — End time (in seconds).

label (*string, optional*) — Optional descriptive label.

Acronyms

JSON JavaScript Object Notation. 4, 5

SALTA Segmentation ALgorithm for Temporal Analysis. 3, 7

XML Extensible Markup Language. 5

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